1. Test Planning and Control

Test Planning

- Specifies how the test strategy and project **Test Plan**

  A document describing the scope, approach, resources and schedule of intended test activities

  apply to the software under test
- Principally:
  - verify the mission,
  - define the Test objectives
  - Specify the Test Activities required to meet the mission and objectives

1. Test Planning and Control

Test Planning (continued)

- Major Tasks are :-
  - Identify the objectives of testing
  - Determine Scope
  - Determine the Test Approach
  - Determine the required test resources
  - Implement the test policy and/or the test strategy
  - Schedule test analysis and design tasks
  - Schedule test implementation, execution and evaluation
  - Determine the Start and Exit Criteria

Jobs for a test manager!!

1. Test Planning and Control

Test Control

- The ongoing activity of comparing actual progress against the plan
- Reporting status, including deviations from the plan
- Taking actions necessary to meet the mission and objectives of the project
- Test Planning takes into account the feedback from monitoring and control activities.
- Major Tasks are :-
  - measure and analyse results
  - Monitor and document progress, test coverage and exit criteria
  - initiate corrective actions
  - make decisions

1. Test Planning and Control

Unit Testing Entry Criteria Example

- Business Requirements are at least 80% complete and have been approved to-date
- Technical Design has been finalized and approved
- Development environment has been established and is stable
- Code development for the module is complete

1. Test Planning and Control

Unit Testing Exit Criteria Example

- Code has version control in place
- No known major or critical defects prevents any modules from moving to System Testing
- A testing transition meeting has be held and the developers signed off
- Project Manager approval has been received

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2. Test Analysis and Design

- Reviewing the test basis (such as requirements, architecture, design, interfaces).
- Evaluating testability of the test basis and test objects.
- Identifying and prioritizing test conditions based on analysis of test items, the specification, behaviour and structure.
- Designing and prioritizing test cases.
- Identifying necessary test data to support the test conditions and test cases.
- Designing the test environment set-up and identifying any required infrastructure and tools.

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2. Test Analysis and Design

- Each Test Case is specified in terms of :-
  - its objective
  - the initial state of the system
  - the input
  - the expected result.

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2. Test Analysis and Design

Fundamental Test Process

Testing Pearl of Wisdom



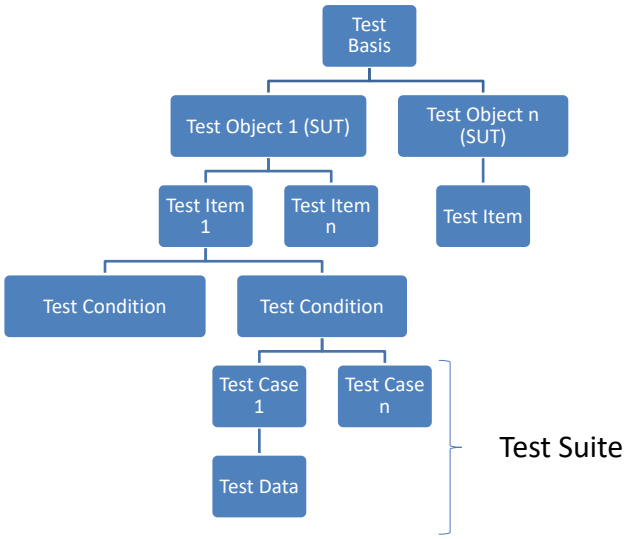
- ***"The act of designing tests is one of the most effective error prevention mechanisms known***

  ... The thought process that must take place to create useful tests can discover and eliminate problems at every stage of development."

Beizer 1983

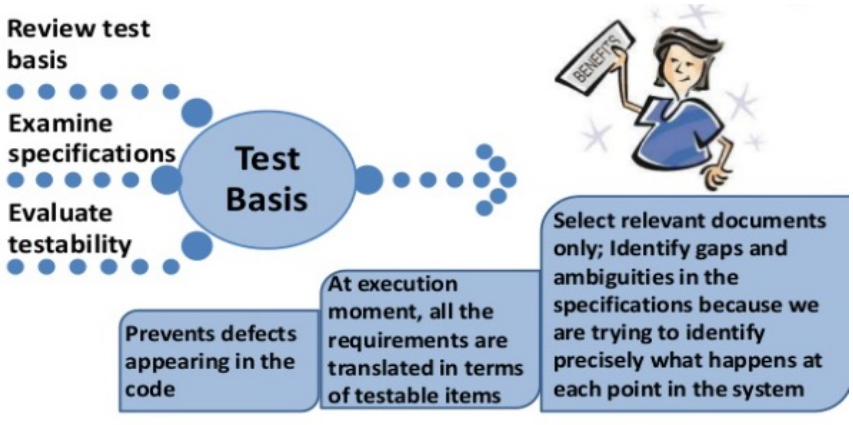
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Test Analysis and Design Vocab



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Important Terms



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Important Terms

Analysis of test items

Identify test conditions

Test Conditions



Gives us a high-level list of what we are interested in testing

We can start identifying the type of generic test data we might need

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Important Terms

Design the tests

Use test design techniques

Test Cases



The high risks areas will be covered by tests before the actual execution phase starts

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Important Terms

Identify test data

Test Data



At execution moment, the test cases will be executed using the most closest data than the one in production

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Important Terms

Design the environment set-up

Identify any infrastructure and tools

Availability

Test Environment



At execution moment, everything we need to carry out our work is in place

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3. Test Implementation and Execution

- Test Cases are combined in a particular order to form **Test Procedures Specifications** (aka Test Scripts) A document specifying a sequence of actions for the execution of a test. Also known as test script or manual test script
- Test Data is created
- The test environment is created
- Major tasks are:

  Create Test Scripts and Data
  - Develop, implement and and prioritise Test Cases
  - Create the Test Scripts
  - Create test data
  - Preparing test harnesses
  - Write automated test scripts
  - Create test suites from the test cases for efficient test execution

  Check the Environment
  - Verify that the test environment has been set up correctly

3. Test Implementation and Execution

Test Script Example

```
def sample_test_script (self):
    type ("TextA")
    click (ImageButtonA)
    assertExist (ImageResultA)
```

3. Test Implementation and Execution

- Major tasks (continued):

Execute the Tests

- Execute the Test Cases (manually or through tools) according to the planned sequence.
- Log the outcome of test execution
- Test execution records should uniquely identify the versions of the software under test, test tools and Testware
- It should be possible to establish that all testing has been carried out by reference to the test records.

3. Test Implementation and Execution

- Major tasks (continued):

Check the Results

- Compare actual results with expected results
- Report discrepancies as Incidents
- Analyse Incidents to establish Root cause
- Repeat, as necessary, test activities as result of action taken for each discrepancy
- The test coverage levels achieved for those measures specified as test completion criteria should be recorded.



3. Test Implementation and Execution

The Execution Phases

- Pre-test / Smoke test:
  - Run a few test cases in order to see if the product is ready for testing.
- Main test:
  - Run all planned test cases.
- After test:
  - Run extra test cases.
  - Repeat the test after repair (regression test).

3. Test Implementation and Execution

Fundamental Test Process



Testing Pearl of Wisdom

- ***"Thoroughly inspect the results of each test"***

Myers - 2004

Ref: Myers, The Art of Software Testing, J Wiley and Sons, 1979

4. Evaluating Exit Criteria and Reporting

- Test execution is assessed against the objectives defined in Test Planning
- This should be done for each **Test Level** (i.e. test stage)

  A group of test activities that are organized and managed together.
- Major tasks are:
  - Check test logs against the exit criteria specified in Test Planning
  - If the exit criteria has not been met
    - Assess if more tests are needed
    - Assess which test activities may need to be repeated
  - Assess if the exit criteria specified should be changed
  - Produce a test summary report for stakeholders review

5. Test Closure Activities

- Collect data from completed test activities to consolidate experience, Testware, facts and numbers
- Major tasks are:
  - Check which planned deliverables have been delivered
  - Check that Incident reports status are up-to-date (e.g. Closed)
  - Ensure all Incident reports have associated change records
  - Record acceptance of the system
  - Finalise and archive Testware, the test environment and the test infrastructure for later reuse
  - Handover Testware to the maintenance organization
  - Analyse lessons learned for future releases and projects, and for process improvement.